

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 05

DATE : 23-06-2019

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 05

Topics

DATE : 23 - 06 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [AIIMS]

PHYSICS : Electrostatics, Gravitation, Current Electricity (Till Ohm's Law), Center of Mass, Collision

CHEMISTRY : Organic Chemistry (Till Alcohol, Phenol & Ether), State of Matter, Chemical Bonding

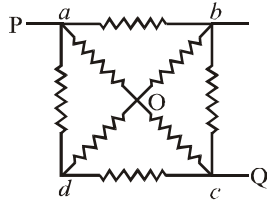
BIOLOGY : Mendalian Genetics (Complete), Molecular Genetics (Up to DNA Replication), Plant Kingdom, Reproductive Health, Skeletal System

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

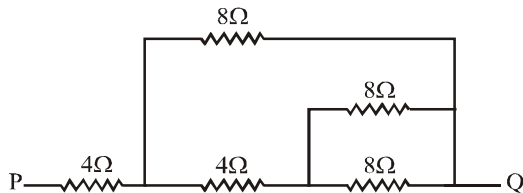
BEWARE OF NEGATIVE MARKING

1. Calculate the resistance between P and Q in the following network. Each element has resistance 3Ω .



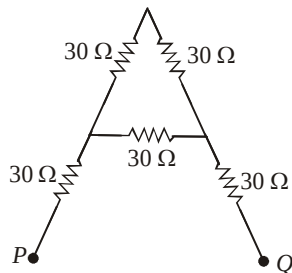
- (1) 6Ω (2) 4Ω
(3) 3Ω (4) 2Ω

2. The resistance between P and Q in the following combination of resistances is



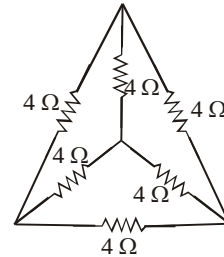
- (1) 8Ω (2) 6Ω
(3) 5Ω (4) 4Ω

3. The effective resistance between points P and Q is



- (1) 150Ω (2) 120Ω
(3) 90Ω (4) 80Ω

4. Six resistances, each equal to 4Ω are connected as shown in the figure. The effective resistance between any two vertices is

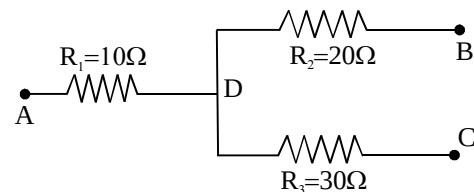


- (1) 2Ω (2) 8Ω
(3) 12Ω (4) 16Ω

5. If a wire is stretched so that its length increases by 0.1% , the percentage change in its resistance will be

- (1) 0.1% (2) 0.2%
(3) 0.3% (4) 0.4%

6. In the circuit shown, $R_1 = 10\Omega$, $R_2 = 20\Omega$ and $R_3 = 30\Omega$. The potentials of points A, B and C are $10V$, $6V$ and $5V$ respectively. The current through resistance R_1 is



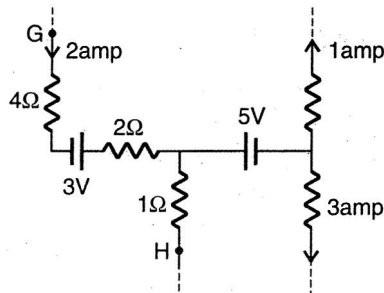
- (1) $0.1A$ (2) $0.2A$
(3) $0.3A$ (4) $0.4A$

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7. Twelve wires, each of resistance 6Ω are connected to form a network of skelton cube. A current enters at one corner and leaves from the diagonally opposite corner. Find the effective resistance between the opposite corners.

(1) 5 Ohm (2) 6 Ohm
(3) 7 Ohm (4) 8 Ohm

8. In the part of a circuit shown in the figure the potentia difference ($V_G - V_H$) between points G and H will be:



(1) 0 V (2) 15 V
(3) 7V (4) 3V

9. By increasing the temperature, the conductance of a conductor and a semiconductor :-

(1) increases for both
(2) decreases for both
(3) increases, decreases
(4) decreases, increases

10. The current flowing through a wire depends on time as $I = 3t^2 + 2t + 5$. The charge flowing through the cross-section of the wire from time $t = 0$ to $t = 2$ sec, is :-

(1) 22 C (2) 20 C
(3) 18 C (4) 5 C

11. Resistance of a wire at 20°C is 20Ω and at 500°C is 60Ω . At what temperature its resistance is 25Ω

(1) 160°C (2) 250°C
(3) 100°C (4) 80°C

12. The resistivity of a wire :-

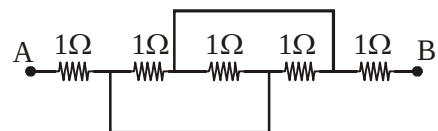
(1) Increases with the length of the wire
(2) Decreases with the area of cross-section
(3) Decreases with the length and increases with the cross-section of wire
(4) None of the above statement is correct

13. Two wires of equal diameters, of resistivities ρ_1 and ρ_2 and length l_1 and l_2 , respectively, are joined in series. The equivalent resistivity of the combination is :-

(1) $\frac{\rho_1 l_1 + \rho_2 l_2}{l_1 + l_2}$ (2) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 - l_2}$

(3) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 + l_2}$ (4) $\frac{\rho_1 l_1 - \rho_2 l_2}{l_1 - l_2}$

14. Equivalent resistance between the point A and B is (in Ω)

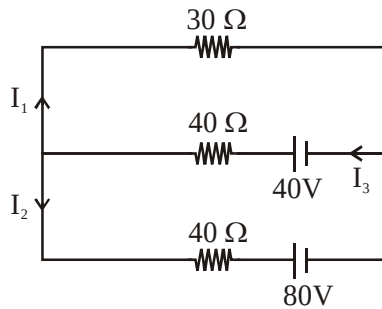


(1) $\frac{1}{5}$ (2) $1\frac{1}{4}$

(3) $2\frac{1}{3}$ (4) $3\frac{1}{2}$

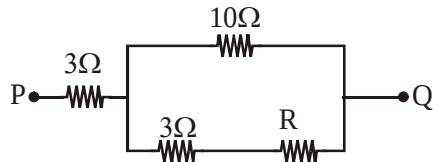
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15. In the given circuit the current I_1 is :-



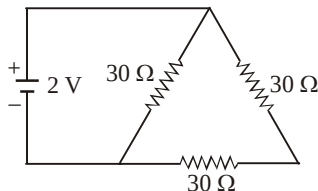
- (1) 0.4 A (2) 1.2 A
(3) 0.8 A (4) -0.8 A

16. In the circuit shown here, what is the value of the unknown resistor R so that the total resistance of the circuit between point P and Q is also equal to R :-



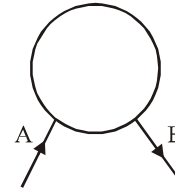
- (1) 3 ohm (2) $\sqrt{39}$ ohm
(3) $\sqrt{69}$ ohm (4) 10 ohm

17. The current in the circuit in the figure is



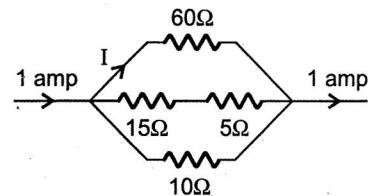
- (1) $\frac{1}{5}$ amp. (2) $\frac{1}{10}$ amp.
(3) $\frac{1}{15}$ amp. (4) $\frac{1}{30}$ amp.

18. A uniform wire of resistance 20Ω having resistance $1 \Omega/\text{m}$ is bent in the form of a circle as shown in the figure. If the equivalent resistance between A and B is 1.8Ω , the length of the shorter section is



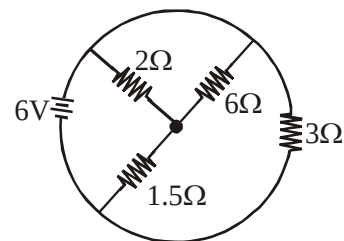
- (1) 1.8 m (2) 2.0 m
(3) 2.5 m (4) 4 m

19. The magnitude of I (in ampere) is :-



- (1) 0.1 (2) 0.3
(3) 0.6 (4) none of these

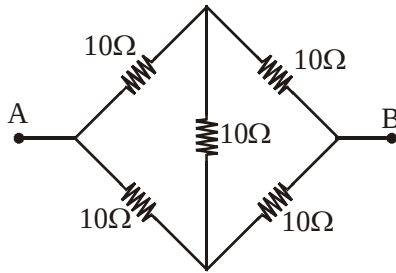
20. The total current supplied to the circuit by the battery is :-



- (1) 1 A (2) 2 A
(3) 4 A (4) 6 A

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21. The effective resistance between points A and B is :-



- (1) $10\ \Omega$
(2) $20\ \Omega$
(3) $40\ \Omega$
(4) None of the above three values
22. A body of mass m rises to a height $h = R/5$ from the earth's surface where R is earth's radius. If g is acceleration due to gravity at the earth's surface, the increase in potential energy is
(1) $m g h$
(2) $(4/5) m g h$
(3) $(5/6) m g h$
(4) $(6/7) m g h$
23. If g is the acceleration due to gravity at the surface of the earth, then the energy required to launch a satellite of mass m from the surface of the earth into a circular orbit at an altitude of $2R$, R being the radius of the earth, is
(1) $\frac{mgR}{6}$
(2) $\frac{mgR}{3}$
(3) $\frac{2mgR}{3}$
(4) $\frac{5mgR}{6}$

24. Acceleration due to gravity on the surface of the moon is $(1/6)$ th of that on the surface of the earth and the diameter of the moon is $(1/4)$ th of that of the earth. If the escape velocity from the earth is 11 km/s then the escape velocity from the moon is
(1) $\frac{11}{24}\text{ km/s}$
(2) $11\sqrt{24}\text{ km/s}$
(3) $\frac{11}{\sqrt{24}}\text{ km/s}$
(4) $11 \times 24\text{ km/s}$
25. If the distance between the earth and the sun were half its present value, the number of days in a year would have been
(1) 64.5
(2) 129
(3) 182.5
(4) 730
26. An artificial satellite moving in a circular orbit around the earth has a total kinetic and potential energy equal to E_0 . Its potential energy is
(1) $-E_0$
(2) $1.5 E_0$
(3) $2 E_0$
(4) E_0
27. A planet has twice the density of earth but the acceleration due to gravity on its surface is exactly the same as on the surface of earth. Its radius in terms of radius of earth R will be
(1) $R/4$
(2) $R/2$
(3) $R/3$
(4) $R/8$
28. Two identical copper spheres each of radius R are placed in contact with each other. If the gravitational attraction between them is F , then
(1) $F \propto R^2$
(2) $F \propto R^4$
(3) $F \propto 1/R^2$
(4) $F \propto 1/R^4$

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29. An infinite number of masses, each of 1 kg, are placed on the positive x-axis at 1 m, 2 m, 4 m, 8 m,.....from the origin. The magnitude of the gravitational field at the origin due to this distribution of masses is
- (1) 2G (2) $\frac{4G}{3}$
- (3) $\frac{3G}{4}$ (4) ∞
30. If a planet has a mass twice that of the earth and its radius is three times that of the earth, then a 1 kg mass on its surface would weigh about
- (1) 1.1 N (2) 2.2 N
- (3) 3.3 N (4) 4.4 N
31. The radius of the earth is 6400 km and $g = 10 \text{ m/s}^2$. In order that a body of 5 kg weighs zero at the equator, the angular speed of the earth should be
- (1) $\frac{1}{180} \text{ rad/s}$
- (2) $\frac{1}{400} \text{ rad/s}$
- (3) $\frac{1}{800} \text{ rad/s}$
- (4) $\frac{1}{1600} \text{ rad/s}$
32. At what depth below the surface of the earth is the value of g same as that at a height of 5 km ?
- (1) 10 km
- (2) 7.5 km
- (3) 5 km
- (4) 2.5 km
33. Two bodies of masses 10^3 kg and 10^5 kg are separated by a distance of 2 m. At what distance from the smaller body will the intensity of the gravitational field be zero ?
- (1) $\frac{1}{11} \text{ m}$ (2) $\frac{2}{11} \text{ m}$
- (3) $\frac{1}{9} \text{ m}$ (4) $\frac{2}{9} \text{ m}$
34. A body of mass 1 kg, which was initially at rest, explodes and breaks into three fragments of masses in the ratio of 1 : 1 : 3. Both the pieces of equal masses fly off perpendicular to each other with a speed of 30 m/s each. The velocity of the heavier fragment is
- (1) $\frac{10}{\sqrt{2}} \text{ m/s}$
- (2) $10\sqrt{2} \text{ m/s}$
- (3) 20 m/s
- (4) $20\sqrt{2} \text{ m/s}$

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| <p>35. If the linear momentum of a body is increased by 50%, its kinetic energy will increase by</p> <p>(1) 50%</p> <p>(2) 100%</p> <p>(3) 125%</p> <p>(4) 150%</p> <p>36. Two perfectly elastic particles A and B of equal masses travelling along the line joining them with velocity 25 m/s and 20 m/s respectively collide. Their velocities after the elastic collision will be (in m/s) respectively</p> <p>(1) 0 and 45</p> <p>(2) 5 and 45</p> <p>(3) 20 and 25</p> <p>(4) 25 and 20</p> <p>37. A 1 kg ball, moving at 12 m/s, collides head-on with a 2 kg ball moving in the opposite direction at 24 m/s. If the coefficient of restitution is $\frac{2}{3}$, then the energy lost in the collision is</p> <p>(1) 60 J</p> <p>(2) 120 J</p> <p>(3) 240 J</p> <p>(4) 480 J</p> | <p>38. A ball is dropped from a height h on the ground. If the coefficient of restitution is e, the height to which the ball goes up after it rebounds for the n^{th} time is</p> <p>(1) he^{2n}</p> <p>(2) he^n</p> <p>(3) $\frac{e^{2n}}{h}$</p> <p>(4) $\frac{h}{e^{2n}}$</p> <p>39. Two particles of masses M and $2M$ are at a distance D apart. Under the mutual gravitational force they start moving towards each other. The acceleration of their centre of mass when they are $D/2$ apart is :</p> <p>(1) $2 GM/D^2$</p> <p>(2) $4 GM/D^2$</p> <p>(3) $8 GM/D^2$</p> <p>(4) Zero.</p> <p>40. A particle is projected vertically upwards with a velocity $\sqrt{gR}$, where R denotes the radius of the earth and g the acceleration due to gravity on the surface of the earth. Then the maximum height ascended by the particle is</p> <p>(1) $R/2$</p> <p>(2) R</p> <p>(3) $2R$</p> <p>(4) $5R/4$</p> |
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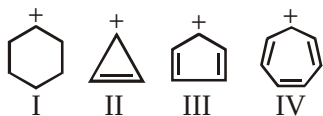
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| <p>41. The hybridization of S in SO_2 is :-
(1) sp (2) sp^2 (3) sp^3 (4) dsp^2</p> <p>42. The shape of a molecule which has 3 σ bond pairs and two lone pair of electrons is :-
(1) tetrahedral
(2) T shape
(3) trigonal bipyramidal
(4) triangular planar</p> <p>43. The types of bonds present in NH_4Cl are :-
(1) electrovalent, covalent and coordinate
(2) only ionic
(3) only covalent
(4) covalent and coordinate</p> <p>44. The linear structure is assumed by :
A : SnCl_2 B : NCO^-
C : NO_2^+ D : CS_2
(1) A, B and C (2) B, C and D
(3) A, C and D (4) B and D</p> <p>45. Which of the following is true?

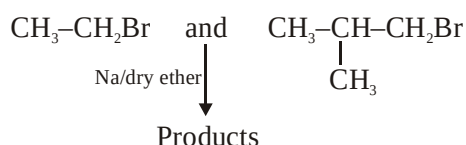
(1) Bond order $\propto \frac{1}{\text{Bond length}} \propto$ bond energy
(2) Bond order $\propto$ bond length $\propto$ bond energy
(3) Bond order $\propto \frac{1}{\text{bond length}} \propto \frac{1}{\text{bond energy}}$
(4) None of these</p> <p>46. Select one correct statement. In the gas equation, $PV=nRT$
(1) n is the number of molecules of a gas
(2) V denotes volume of one mole of the gas
(3) n moles of the gas have a volume V
(4) P is the pressure of the gas when only one mole of gas is present.</p> | <p>47. A gaseous mixture of 2 moles of A, 3 moles of B, 5 moles of C and 10 moles of D is contained in a vessel. Assuming that the gases are ideal and the partial pressure of C is 1.5 atm, the total pressure is
(1) 3 atm (2) 6 atm
(3) 9 atm (4) 15 atm</p> <p>48. The ratio of rates of diffusion of SO_2, O_2 and CH_4 is
(1) $1:\sqrt{2}:2$ (2) $1:2:4$
(3) $2:\sqrt{2}:1$ (4) $1:2:\sqrt{2}$</p> <p>49. The compressibility factor of a gas is less than unity at STP. Therefore,
(1) $V_m > 22.4 \text{ L}$ (2) $V_m < 22.4 \text{ L}$
(3) $V_m = 22.4 \text{ L}$ (4) $V_m = 44.8 \text{ L}$</p> <p>50. The relationship between P_c, V_c and T_c is
(1) $P_c V_c = RT$
(2) $P_c V_c = 3RT_c$
(3) $P_c V_c = \frac{3}{5}RT_c$
(4) $P_c V_c = \frac{3}{8}RT_c$</p> <p>51. Most acidic compound is :
(1) CH_3COOH
(2) $\text{C}_6\text{H}_5\text{COOH}$
(3) $\text{O}_2\text{NC}_6\text{H}_4\text{CO}_2\text{H}$
(4) $\text{C}_6\text{H}_5\text{OH}$</p> <p>52. Maximum $-I$ effect is exerted by the group :
(1) C_6H_5 (2) OCH_3
(3) Cl (4) NO_2</p> |
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53. The stability of the following carbocation decrease in the order :



- (1) IV > III > II > I (2) IV > II > III > I
(3) IV > II > I > III (4) IV > I > II > III
54. Which of the following is not prepared by Kolbe's electrolytic process ?
(1) C_3H_8 (2) C_2H_6
(3) C_4H_{10} (4) C_6H_{14}
55. Number of di-substituted isomers of the product of the reaction,
 $CH_3CH_2CH_3 + Br_2 \xrightarrow{h\nu}$ would be :
(1) 2 (2) 1 (3) 3 (4) 4
56. What is the maximum number of carbon atoms in the expected products of the Wurtz reaction?



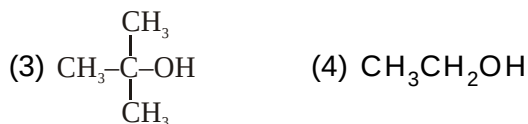
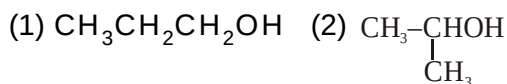
- (1) 8 (2) 6 (3) 4 (4) 2
57. Grignard reagent C_2H_5MgBr gives alkane with :
(1) H_2O (2) C_2H_5OH
(3) $C_2H_5NH_2$ (4) all of these
58. Kolbe's synthesis of sodium salt of butanoic acid gives:
(1) n-hexane (2) iso-butane
(3) n-butane (4) propane
59. When propanal is heated with Zn-Hg and conc. HCl, which is formed ?
(1) C_3H_4 (2) C_3H_6
(3) C_3H_8 (4) C_3H_7Cl

60. X is heated with sodalime and gives ethane. X is :

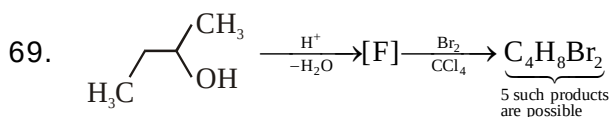
- (1) ethanoic acid (2) methanoic
(3) propanoic acid (4) 2-methyl pentane
61. 1-Chlorobutane when treated with alcoholic potash gives :
(1) 1-butene (2) 1-butanol
(3) 2-butene (4) 2-butanol
62. The nature of hydrogen atom in chloroform molecule is :
(1) neutral
(2) acidic
(3) basic
(4) cannot be predicted
63. Which of the following alkyl halide is hydrolysed by S_N1 mechanism ?
(1) $(CH_3)_2CHX$ (2) CH_3CH_2X
(3) $CH_3CH_2CH_2X$ (4) $(CH_3)_3CX$
64. $(CH_3)_2CHCl + NaI \xrightarrow{\text{Acetone}} (CH_3)_2CHI + NaCl$
The above reaction is known as :
(1) Finkelstein reaction
(2) Stephen's reaction
(3) Kolbe's reaction
(4) Wurtz reaction
65. The non-reactivity of chlorine atom in $H_2C=CH-Cl$ is due to :
(1) inductive effect
(2) resonance stabilization
(3) electromeric effect
(4) electronegativity
66. $(X) + KCN \rightarrow CH_3CN \xrightarrow{2H_2/Ni} CH_3CH_2NH_2$
what is (X) ?
(1) CH_3CH_2Cl (2) CH_3Cl
(3) $CH_3CH_2CH_2Cl$ (4) $(CH_3)_2CHCl$

SPACE FOR ROUGH WORK

67. Which one of the following alcohols reacts most readily with Lucas reagent ?



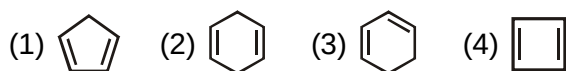
68. The product of reaction $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[\text{Cu}]{300^\circ\text{C}} ?$ is :



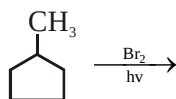
How many isomers of [F] are possible ?



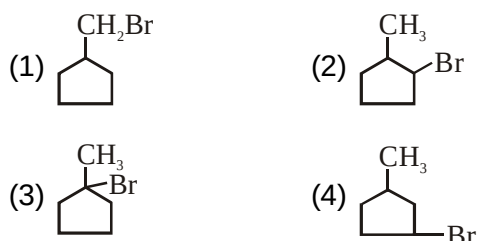
70. Ozonolysis products of an olefin are $\text{OHC}-\text{CHO}$ and $\text{OHC}-\text{CH}_2-\text{CH}_2-\text{CHO}$, the olefin is :



71. In the following reaction is :



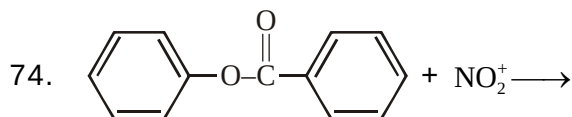
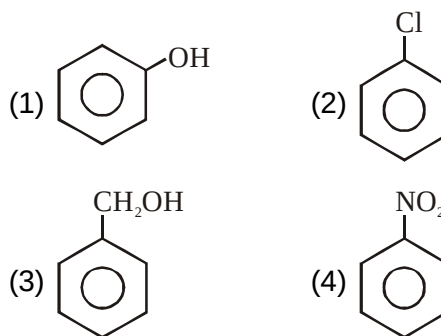
The major product obtained :



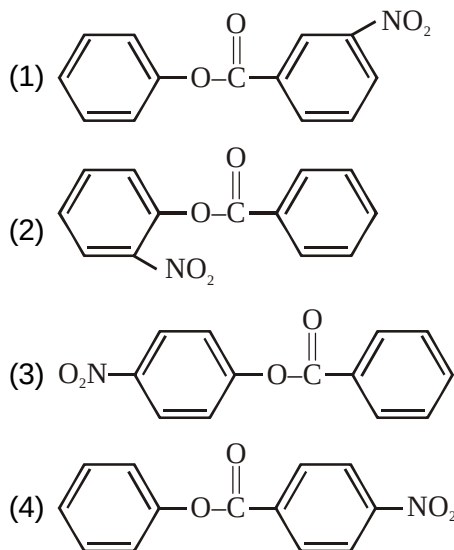
72. The reaction between ethylene and bromine is an example of :-

- (1) electrophilic addition
(2) electrophilic substitution
(3) nucleophilic addition
(4) nucleophilic substitution

73. Which one of the following is most reactive towards electrophilic attack?



Product of this reaction by single (S_E) electrophilic substitution is :-



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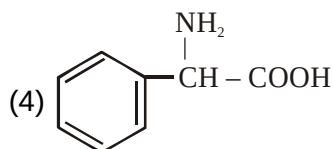
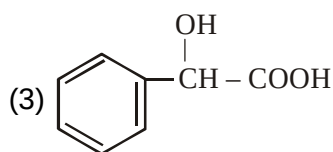
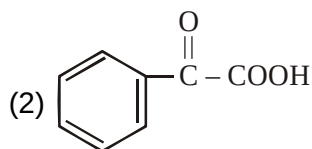
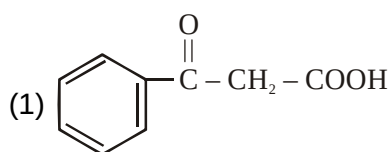
75. NaBH_4 can reduce which of the following compound?

- (1) $\text{CH}_3\text{CH}_2 - \text{OH}$ (2) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{Cl}$
 (3) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{OH}$ (4) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{NH}_2$

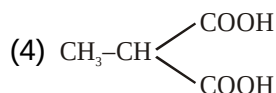
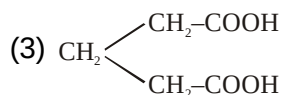
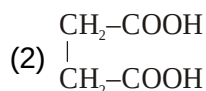
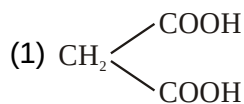
76. Benzene is a polymer of :-

- (1) methane (2) acetylene
 (3) ethane (4) ethylene

77. Which of the following carboxylic acids undergoes decarboxylation most easily?



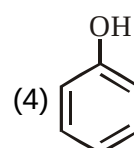
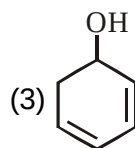
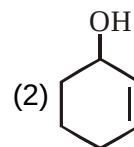
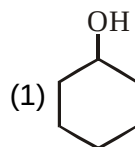
78. A dibasic organic acid (A) on heating gives (B). The compound (B) on soda lime treatment gives the lowest alkane which can be prepared by Wurtz synthesis. The compound (A) is :-



79. The correct decreasing order of preference of functional groups during the IUPAC nomenclature of polyfunctional compounds is:

- (1) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$
 (2) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$
 (3) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
 (4) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$

80. Which of the following will be most easily dehydrated in acidic conditions :-



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- 81.** Identify the correctly matched pair.

 - (1) $\phi \times 174$ – 5836 Nucleotide
 - (2) Bacteriophage – 48502 BP
 - (3) *E. Coli*- 4.6×10^{12} BP
 - (4) Human genome – 3.3×10^6 BP

82. In the B-DNA, number of nitrogenous base pairs in 4 turn:

 - (1) 40
 - (2) 48
 - (3) 20
 - (4) 44

83. In the bacteriophage $\phi \times 174$ number of nucleotide presents are 5386. Calculate the number of phospho-diester bond and pentose sugar in it.

 - (1) 5385, 5386
 - (2) 5386, 5386
 - (3) 5386, 5385
 - (4) 5385, 5385

84. Which of the following findings are not associated with work of Chargaff on DNA?

 - (1) $\frac{C}{A} = 1$, follow chargaff rule
 - (2) $\frac{A+G}{T+C} = 1$, this value is constant for all species
 - (3) Base pair ratio can be used for identification of species
 - (4) Purines and pyrimidines occur in equal amount in one strand

85. Consider the following statements and choose the correct option stating true (T) or False (F).

 - (A) Compaction occur by folding and attachment of DNA with polyamines in prokaryotes
 - (B) The compact structure of DNA is called nucleosome in prokaryotes.
 - (C) Histone proteins are rich in basic amino acid.
 - (D) 'Nu body' consist of two copies of each H_2A , H_2B , H_3 and H_4 histone protein.
 - (E) Length of DNA for a human diploid cell is 2.2 metre.

	A	B	C	D	E
(1) T		F	T	T	T
(2) F		F	T	F	T
(3) F		T	T	T	F
(4) T		T	T	T	T

86. The unequivocal proof that DNA is genetic material came from the experiment of:

 - (1) Hershey and chase
 - (2) Meselson and Stahl
 - (3) Avery, Macleod and Mccharty
 - (4) Griffith

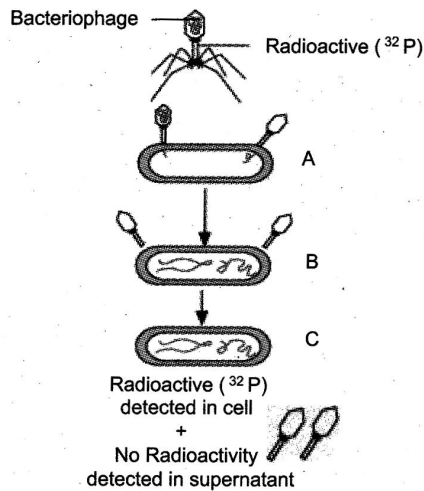
87. Consider the following statements and give your correct conclusion.

 - (A) DNA is chemically less reactive and structurally more stable than RNA.
 - (B) Presence of thymine at place of uracil confers additional stability to DNA.
 - (C) RNA can directly code for the synthesis of proteins hence can easily express the characters.
 - (D) DNA is dependent on RNA for protein synthesis.

Options are :

 - (1) Only DNA act as genetic material
 - (2) RNA is better material for transmission of information
 - (3) Only DNA can act as genetic material in prokaryotes and RNA can act as genetic material in Eukaryotes
 - (4) DNA is an adapter, structural molecule and in some cases it is catalytic

88. Identify the stages in the given figure.



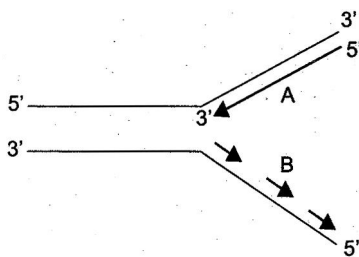
A B C

- (1) Infection Blending Centrifugation
(2) Blending Infection Centrifugation
(3) Infection Centrifugation Blending
(4) Centrifugation Blending Infection

89. The enzyme that removes the RNA primer after DNA replication is:

- (1) Primase
(2) DNA - polymerase I
(3) DNA polymerase II
(4) Ligase

90. Choose the **odd** one w.r.t. given figure.



- (1) Figure shows that DNA replication occur within replication fork
(2) Helicase enzyme cuts hydrogen bond between two strands and results in formation of Y-shaped structure
(3) 'B' is lagging strand and synthesised in $3' \rightarrow 5'$ direction
(4) The parent DNA template with polarity $3' \rightarrow 5'$ has continuous replication

91. Read the following statements carefully w.r.t. DNA replication:

- (i) The main enzyme is DNA dependent DNA polymerase.
(ii) Deoxyribonucleoside monophosphates provides energy for polymerisation.
(ii) Polymerisation occur only in one direction that is $5' \rightarrow 3'$.
(iv) Prokaryotes have single 'ori' site.
(v) In Eukaryotes, the replication of DNA takes place at S phase of cell cycle.

How many of the above given statements are correct?

- (1) Four
(2) Two
(3) Three
(4) One

92. Match the following and choose the correct combination from the options given:

Column I

Column II

- A. $Rr \times Rr$ 1. 1 : 1
B. $Rr \times rr$ 2. 3 : 1
C. $RrYy \times RrYy$ 3. 1 : 1 : 1 : 1
D. $RrYy \times rryy$ 4. 9 : 3 : 3 : 1

- (1) A = 2, B = 4, C = 3, D = 1
(2) A = 1, B = 3, C = 2, D = 4
(3) A = 4, B = 3, C = 2, D = 1
(4) A = 2, B = 1, C = 4, D = 3

93. Find out phenotypic categories in following cross-

A – $AaBbCC \times aaBBCc$

B – $GgRrTt \times ggrTT$

C – $aabbCC \times AABbCc$

	A	B	C
(1)	2	4	1
(2)	2	2	2
(3)	2	3	1
(4)	3	2	2

94. In a plant species, flower colour yellow dominant over white and fruit shape round is dominant over elongated. Crossing was performed between two pure lines one having yellow flower and round fruit and another with white flower and elongated fruit. About 20 plants survived in F_1 progeny. All plant of F_1 were allowed to selffertilize and about 960 plants survived in F_2 . If the traits follow Mendelian inheritance the number of plant would have yellow flower and round fruit in F_1 and F_2 are respectively
- (1) 20, 960
 - (2) 20, 540
 - (3) 10, 180
 - (4) 10, 60
95. Mendel's principle of segregation means that gametes always receive :-
- (1) One pair of alleles
 - (2) One quarter of the genes
 - (3) One of the paired alleles
 - (4) Any pair of alleles
96. Sporophyte dependent upon gametophyte is found in-
- (1) Algae
 - (2) Fungi
 - (3) Bryophyta
 - (4) Pteridophytes
97. Fern prothallus is developed from-
- (1) Spore
 - (2) Oospore
 - (3) Elater
 - (4) Antherozoid
98. Selaginella and salvinia are the plants of
- (1) Homosporous, Hetrosporous
 - (2) Hetrosporous, Homosporous
 - (3) Hetrosporous, Hetrosporous
 - (4) Homosporous, Homosporous
99. Bryophytes are dependent on water because-
- (1) Archegonium has to remain filed with water for fertilization
 - (2) Water is non-essential for fertilization for life
 - (3) Water is essential for their vegetative progagation
 - (4) The sperms can easily reach upto egg in the archegonium
100. Seed of Pinus shows three generations as-
- (1) Parent sporophyte, gametophyte and future sporophyte
 - (2) Parent gametophyte, sporophyte and future gametophyte
 - (3) Parent sporophyte, sporophyte, future gametophyte
 - (4) None of these

101. In India, the family planning program was launched in:

- (1) 1977 (2) 1951
(3) 1967 (4) 1981

102. Prenatal genetic defects in foetus can be detected by:

- (1) CT-scan
(2) X-ray radiography
(3) MRI
(4) Amniocentesis

103. Which of the following is the cause of population explosion in India?

- (1) Rapid decline in death rate
(2) Decline in mortality maternal rate (MMR)
(3) Increase number of people in reproductive age
(4) All of these

104. Emergency contraceptives are effective if used within:

- (1) 72 hours of fertilization
(2) 72 hours of coitus
(3) 72 hours of implantation
(4) 72 hours of ovulation

105. Match the contraceptive methods given under Column I with their examples given under Column II. Select the correct choice from those given below.

Column-I		Column-II	
A.	Barrier device	p.	Depo-Provera
B.	Sterilization	q.	Condom, Diaphragm
C.	IUDs	r.	Multiload-375
D.	Injectable contraceptive	s.	Vasectomy

- (1) A-q, B-p, C-r, D-s
(2) A-f, B-s, C-p, D-q
(3) A-q, B-s, C-r, D-p
(4) A-s, B-q, C-p, D-r

106. Zygote or early embryo upto eight blastomeres is transferred into fallopian tube by a procedure called:

- (1) Intra Uterine Transfer (IUT)
(2) Zygote Intra Fallopian Transfer (ZIFT)
(3) Gamete Intra Fallopian Transfer (GIFT)
(4) Intra Uterine Insemination (IUI)

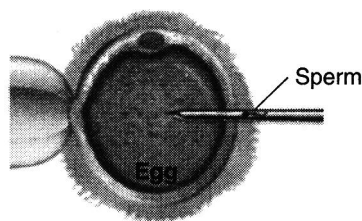
107. Which of the following statement is correct regarding MTP?

- (1) 60-65 million pregnancies are terminated world wide every year.
(2) One fifth of the total conceived pregnancies are terminated.
(3) MTP can be used for contraception.
(4) MTPs are safe upto 20 weeks.

108. Which of the following is incorrect regarding contraceptive devices?

- (1) Progestasert and LNG-20 are hormone releasing IUDs.
(2) Spermicidal creams and jellies increase the efficiency of barrier devices.
(3) Copper-T is placed in vagina and can be self inserted.
(4) Condoms are useful in protecting against STDs.

109. What is the figure given below showing one of the assisted reproductive technologies?



- (1) ZIFT (2) IUT
(3) GIFT (4) ICSI

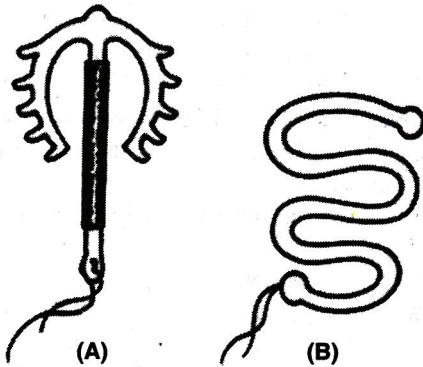
110. Which one of the following statements is wrong?

- (1) The national 'family planning program' was launched in India in 1951.
(2) MTPs are considered relatively safe upto second trimester.
(3) Contraceptive methods are commonly used to prevent unwanted pregnancies.
(4) In India, Nirodh is popularly marketed condom made of latex.

111. Which one of the following statements is incorrect?

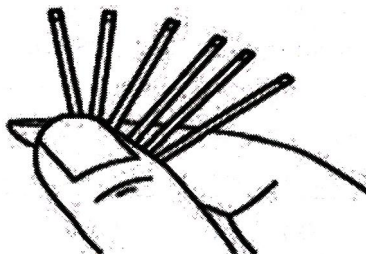
- (1) Medical termination of pregnancy is legalised in our country.
- (2) Coitus interruptus is one of the oldest forms of contraceptive methods known to man.
- (3) *In vitro* fertilization (IVF) is a simple, cost-effective procedure for infertile couples.
- (4) Saheli is the world's first non-steroidal oral contraceptive pills.

112. Two Intra Uterine Devices (IUD) are given below. Correctly identify them by selecting the option:



- (1) A - LNG -20, non-medicated IUD
B - Lippes loop, hormonal IUD
- (2) A - Multiload 375, Cu-releasing IUD
B - Lippes loop, non-medicated IUD
- (3) A - Lippes loop, hormonal IUD
B - Copper T, non-medicated IUD
- (4) A - Lippes loop, hormonal IUD
B - Copper 7, non-medicated IUD

113. Identify the image of the device given below used as capsules surgically implanted under skin slowly release hormone that blocks ovulation?



- (1) Levonorgestrel
- (2) Progestasert
- (3) Norplant
- (4) Depo-Provera

114. Diaphragms are contraceptive devices used by the females. Choose the correct options from the statements given below:

- i. They are introduced into the uterus
 - ii. They are placed to cover the cervical region
 - iii. They act as physical barriers for sperm entry
 - iv. They act as spermicidal agents
- (1) i and ii
 - (2) i and iii
 - (3) ii and iii
 - (4) iii and iv

115. Which of the following is an example of sesamoid bone?

- (1) Pisiform
- (2) Lunette
- (3) Scaphoid
- (4) Cuboid

116. The digital formula of hand is

- (1) 02233
- (2) 22343
- (3) 23333
- (4) 03333

117. Obturator foramen is present between

- (1) Pubis and Ilium
- (2) Pubis and Ischium
- (3) Ilium and Ischium
- (4) None of these

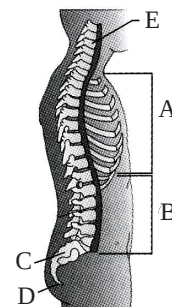
118. The types of joints between human skull bones are

- (1) Saddle
- (2) Fibrous
- (3) Synovial
- (4) Cartilaginous

119. In human beings, cranium is formed by

- (1) Ten bones of which two are paired
- (2) Eight bones of which two are paired
- (3) Twelve bones of which one are paired
- (4) Ten bones of which one is paired

120. Identify the parts labelled A, B, C and D for the given diagram of human vertebral column and select the correct option.



	A	B	C	D	E
(1)	Thoracic Vertebra	Lumbar Vertebra	Sacrum	Coccyx	Cervical Vertebra
(2)	Cervical Vertebra	Lumbar Vertebra	Thoracic Vertebra	Sacrum	Coccyx
(3)	Cervical Vertebra	Thoracic Vertebra	Lumbar Vertebra	Sacrum	Coccyx
(4)	Cervical Vertebra	Thoracic Vertebra	Lumbar Vertebra	Coccyx	Sacrum

Direction for Assertion & Reason questions.
 These questions consist of two statements each, printed as assertion and reason. while answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. **Assertion :** Gravitational force between two particles is negligibly small compared to the electrical force.

Reason : The electrical force is experienced by charged particles only.

122. **Assertion :** The universal gravitational constant is same as acceleration due to gravity.

Reason : Gravitational constant and acceleration due to gravity have same dimensional formula.

123. **Assertion :** If earth suddenly stops rotating about its axis, then the value of acceleration due to gravity will become same at all the places.

Reason : The value of acceleration due to gravity is independent of rotation of earth.

124. **Assertion :** Gravitational potential of earth at every place on it is negative.

Reason : Every body on earth is bound by the attraction of earth.

125. **Assertion :** A planet moves faster, when it is closer to the sun in its orbit and vice versa.

Reason : Orbital velocity in orbital of planet is constant.

126. **Assertion :** The time period of geostationary satellite is 24 hours.

Reason : Geostationary satellite must have the same time period as the time taken by the earth to complete one revolution about its axis.

127. **Assertion :** Two different planets have same escape velocity.

Reason : Value of escape velocity is a universal constant.

128. **Assertion :** A body becomes weightless at the centre of earth.

Reason : As the distance from centre of earth decreases, acceleration due to gravity increases.

129. **Assertion :** Gravitational field is zero both at centre and infinity.

Reason : The dimensions of gravitational field is $[LT^{-2}]$.

130. **Assertion :** The resistivity of a semiconductor increases with temperature.

Reason : The atoms of a semiconductor vibrate with larger amplitude at higher temperatures thereby increasing its resistivity

131. **Assertion :** In a simple battery circuit the point of lowest potential is positive terminal of the battery

Reason : The current flows towards the point of the higher potential as it flows in such a circuit from the negative to the positive terminal.

132. **Assertion :** The temperature coefficient of resistance is positive for metals and negative for *p*-type semiconductor.

Reason : The effective charge carriers in metals are negatively charged whereas in *p*-type semiconductor they are positively charged.

133. **Assertion :** There is no current in the metals in the absence of electric field.

Reason : Motion of free electron are randomly.

134. *Assertion* : The drift velocity of electrons in a metallic wire will decrease, if the temperature of the wire is increased.

Reason : On increasing temperature, conductivity of metallic wire decreases.

135. *Assertion* : Bending a wire does not effect electrical resistance.

Reason : Resistance of wire is proportional to resistivity of material.

136. *Assertion* : The connecting wires are made of copper.

Reason : The electrical conductivity of copper is high.

137. *Assertion* : The centre of mass of a proton and an electron, released from their respective positions remains at rest.

Reason : The centre of mass remain at rest, if no external force is applied.

138. *Assertion* : The position of centre of mass of a body does not depend upon shape and size of the body.

Reason : Centre of mass of a body lies always at the centre of the body.

139. *Assertion* : Location of centre of mass is independent of the reference frame.

Reason : Centre of mass is same as centre of gravity.

140. *Assertion* : In an elastic collision of two billiards balls, the kinetic energy is not conserved during the short interval of time of collision between the balls.

Reason : Energy spent against friction does not follow the law of conservation of mechanical energy.

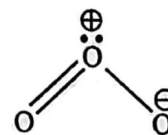
141. *Assertion* : All the P–F bonds in PF_5 do not have equal bond lengths.

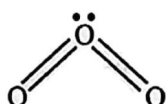
Reason : The hybridisation of P-atom involves not only s and p-orbitals but also d-orbitals.

142. *Assertion* : PCl_5 is not very stable and easily dissociates to form PCl_3 and Cl_2 .

Reason : The central atom (P) in PCl_5 has 10 electrons around it instead of 8, i.e., it violates octet rule.

143. *Assertion* : The electronic structure of O_3 is



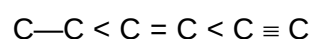
Reason :  structure is not

allowed because octet around O cannot be expanded.

144. *Assertion* : Addition of HBr to 1-butene gives two optical isomers.

Reason : The product contains two chiral carbon atom.

145. *Assertion* : Bond energy has following sequence :



Reason : Bond energy increases with increase in bond order.

146. *Assertion* : Tertiary carbonium ions are generally formed more easily than primary carbonium ions.

Reason : Hyperconjugative as well as inductive effect due to additional alkyl groups stabilize tertiary carbonium ions.

147. *Assertion* : *Trans* isomers of But-2-ene is more stable than *cis* isomer.

Reason : The *cis* isomer is the one in which two similar groups are on the same side of double bond.

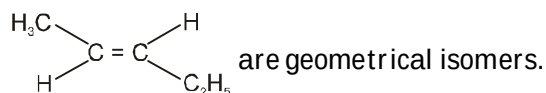
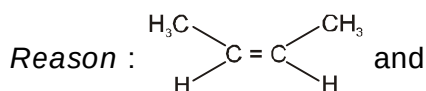
148. *Assertion* : All the hydrogen atoms in $\text{CH}_2 = \text{C} = \text{CH}_2$ lie in one plane.

Reason : All the carbon atoms in it are sp^2 hybridized.

149. *Assertion* : Rotation about $\text{C} = \text{C}$ bond is restricted.

Reason : All compounds having $\text{C} = \text{C}$ bond exhibit geometrical isomerism.

150. **Assertion** : Geometrical isomers have same molecular formula but different orientation in space.



151. **Assertion** : Organic molecules having only 1 chiral carbon atom are always optically active.

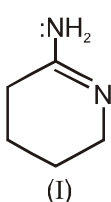
Reason : Presence of chiral carbon is essential condition for any molecules showing optical activity.

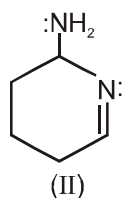
152. **Assertion** : Carbon-oxygen bonds are of equal length in acetate ion.

Reason : Bond length decreases with the multiplicity of bond between two atoms.

153. **Assertion** : Size of double bond in benzene is more than the size of double bond in buta-1,3-diene.

Reason : Increase of delocalisation of π electrons increases the size of double bond.

154. **Assertion** :
- 
- (I) is stronger base than



Reason : The $-\text{NH}_2$ group of (I) is better electron pair donor to H^+ ion than the $-\text{NH}_2$ group of (II)

155. **Assertion** : AgNO_3 increases the rates of solvolyses, So rate of S_N^1 reaction increases.

Reason : Ag^+ has a stronger affinity for X^- , than a solvent molecule, the precipitation of AgCl accelerates the dissociation of X^- so rate of S_N^1 reaction increases.

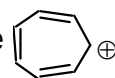
156. **Assertion** : Ether solvent used in Wurtz reaction must be absolutely water free.

Reason : Water reacts with Me-Cl to produce Me-OH

157. **Assertion** : Propane on photochlorination gives 2-chloro propane as major product.

Reason : $\text{Me}-\overset{+}{\text{C}}\text{H}-\text{Me}$ carbocation is more stable than $\text{MeCH}_2\overset{+}{\text{C}}\text{H}_2$ carbocation.

158. **Assertion** : Tropylium cation is aromatic in nature



Reason : The only property that determines its aromatic behaviour is its planar structure.

159. **Assertion** : $\text{Me}_3\overset{+}{\text{C}}$ is more stable than

$\text{Me}_2\overset{+}{\text{C}}\text{H}$ and $\text{Me}_2\overset{+}{\text{C}}\text{H}$ is more stable than the $\text{Me}\overset{+}{\text{C}}\text{H}_2$.

Reason : Greater the number of hyperconjugative structures, more is the stability of carbocation.

160. **Assertion** : CH_3OCH_3 and $\text{C}_2\text{H}_5\text{OH}$ have same molecular masses but boiling point of $\text{C}_2\text{H}_5\text{OH}$ is higher than dimethyl ether.

Reason : $\text{C}_2\text{H}_5\text{OH}$ forms intermolecular H-bonding while CH_3OCH_3 forms intramolecular H-bonding.

161. *Assertion*: In general, Bryophytes are of little economic importance.
Reason : Some mosses provide food for herbaceous organisms. *Sphagnum* species provides peat. *Sphagnum* species is also used in packing material for trans-shipment of living material because of its good water holding capacity.
162. *Assertion* : The sporophyte of liverworts is differentiated into foot, seta and capsule.
Reason : In Riccia, sporophytes are represented by a capsule.
163. *Assertion* : If the sequence of bases in one strand is known then the sequence of the other strand can be predicted.
Reason : Both the strands of DNA are complementary to each other.
164. *Assertion* : There is approximately a uniform distance between the two strands of DNA helix.
Reason : In DNA, always a purine comes opposite to a pyrimidine.
165. *Assertion* : In DNA, histones are organized to form a unit of four molecules called histone tetramer.
Reason : The negatively charged DNA is wrapped around a positively charged histone tetramer to form a structure called nucleosome.
166. *Assertion* : DNA and RNA are the two types of nucleic acids present in living organisms.
Reason : DNA acts as a genetic material in most of the organisms.
167. *Assertion* : To determine the genotype of a tall plant at f_2 , Mendel crossed the tall plant from f_2 , with a dwarf plant (test cross).
Reason : In a typical test cross, an organism showing a dominant phenotype (and whose genotype is to be determined) is crossed with the dominant parent instead of self-crossing.
168. *Assertion* : The progenies of test cross can easily be analyzed to predict the genotype of the test organism.
Reason : Heterozygous test organism produces 1 : 2 : 1 phenotype ratio in test cross.
169. *Assertion* : According to the law of dominance, characters are controlled by discrete units called factors, which occur in pairs.
Reason : In a dissimilar pair of factors, one member of the pair dominated the other.
170. *Assertion* : The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross in the F_1 and the expression of both in the F_2 .
Reason : It explains the proportion of 3 : 1 obtained at F_2 , and also explains the dominance of one trait in F_1 .
171. *Assertion* : Surgical methods block gamete transport and thereby prevent conception.
Reason : Surgical method using in female for this purpose is called vasectomy.
172. *Assertion* : Lactational amenorrhoea refers to absence of menstruation and ovulation during intense lactation following parturition.
Reason : Lactational amenorrhoea is due to high levels of prolactin which inhibits release of gonadotropins.
173. *Assertion* : *In vitro* fertilization refers to fusion of gametes outside the body in similar conditions in laboratory.
Reason : The fertilizing capacity of sperms is for 24-48 hours.
174. *Assertion* : Barrier methods like condoms protect users from sexually transmitted diseases (STDs).
Reason : These devices cover penis in male or vagina and cervix in female and prevent ejaculated semen from getting released in female genital tract.

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| <p>175. <i>Assertion</i> : 'Saheli' is the world's first nonsteroidal oral contraceptive pill.
<i>Reason</i> : It has been developed by Central Drug Research Institute, Lucknow.</p> <p>176. <i>Assertion</i> : IUDs are effective contraceptive devices in females.
<i>Reason</i> : These increase phagocytosis of sperms within uterus.</p> <p>177. <i>Assertion</i> : Barrier methods of contraception are available only for females.
<i>Reason</i> : It prevents the first cleavage divisions of zygote.</p> | <p>178. <i>Assertion</i> : Sterilization procedure in the male is called vasc-tomy.
<i>Reason</i> : In this epididymis is removed preventing sperm maturity and motility.</p> <p>179. <i>Assertion</i> : Government of India legalised MTP in the year 1987.
<i>Reason</i> : It is mainly meant to get rid of unwanted pregnancies.</p> <p>180. <i>Assertion</i> : Human skull is dicondylic.
<i>Reason</i> : Atlas has two condyles using which it rests on vertebral column.</p> |
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